Departamento de Ingeniería Industrial
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Understanding Consumer Behavior through Discrete Choice Models

Homework Assignment No. 2
Multinomial Choice Models – Application to champagne brand choice

Date distributed: April 29, 2025

Due date: Friday May 23, 2025, 11:59PM

The file HW02_data_champ.csv defines the data to be used for this assignment. The data set comprises 2,650 observations. Each record represents an adult (individual who is 18 years or older). This data contains information about a research experiment conducted by a private company in France in 2016. Random individuals were selected and interviewed on their "wine drinking habits". From the whole sample, we have selected here only those individuals who stated being occasional or regular drinkers of champagne (white sparkling wine made in the old province of Champagne, France). In this homework, we will study the choice between four different brands of champagne: Champagne Delamotte (DEL), Champagne Gosset (GOS), Champagne Veuve Clicquot Ponsardin (VCP), and Champagne Henri Mandois (HM). The definitions of the variables in the data set are attached to the end of document.

Notes (very important)

- Consider 95% as the level of confidence for statistically significant parameters ($|t| \geq 1.96$)
- Participants were asked to **rate** (from 1 to 5; 1: very bad, 2: bad, 3: neutral, 4: good, and 5: very good) 6 **attributes** of the different brands. You can consider these rates to enter in the utility function as they are (no need to convert them to dummy variables).
- You can consider that the coefficients associated to alternative characteristics (such as price, **attribute rates**, etc.) do not vary across alternatives, i.e., they are the same for all alternative utility functions.
- All the respondents said they knew the brands beforehand. Prices were given but participants were told to forget budget constraints (after all, they did not have to buy the products).

1. Examine Characteristics of Data Set (15 points)

   Your first task is to become familiar with characteristics of your data set. Please use frequency tables, descriptive statistics, and cross tabulations for this purpose. Report only the most interesting findings of this exploration. Focus your comments only on those items you think are most important or interesting. Do you think the actual population of France is well represented in the sample?

2. Estimate a Multinomial Logit Model (15 points)

   Use what you already know to develop a preferred specification for brand choice with this data. Look for the best specification (in terms of goodness of fit measures) that you can explain intuitively. Briefly discuss the intuition behind the value (sign) of each of the coefficients. You are free to transform the explanatory variables. Use your imagination. The sky is the limit.

3. Market Segmentation Analysis (10 points)

   Consider the preferred specification you obtained in part 2. Use this same specification, but now estimate models for different segment groups based on GENDER. Estimate a fully segmented model and compare this with your preferred specification in part 2. Next, independent of the results of the test, describe any major differences among the estimated segment-level models and interpret these differences. Are the differences reasonable from an intuitive standpoint? Report your results in a table format that includes the specification in part 2 and each segment-level model result.

4. Random coefficients (10 points)

   Now exclude the individuals who has chosen GOS, and focus only on the choice DEL v/s VCP v/s HM. You can use the following (or similar) command: *database = subset(database,database$CHOSEN!=2)*. Test if any of the estimated coefficients in part 2 can be considered random (i.e., distributed normal or log-normal). Only test those coefficients that are associated to **attribute rates**. Use 200 draws for the simulations (no need for more). After performing the tests, update (if necessary) your utility specification.

5. Prediction exercise (10 points)

   Using your preferred specification from part 4, compute the probability of choice of each alternative for each individual and list in a table the top 10 and worst 10 for each alternative, along with the average probability (across the sample) for each alternative. Which brand is generally preferred in the sample? How would you advise the different brand's marketing/design division?

Description of file "HW02_data_champ"

PID	Person identification number
CHOSEN	Chosen brand. 1: DEL, 2: GOS, 3: VCP, and 4: HM.
GENDER	Individual's gender. 1: Woman, 2: Man.
AGE	Individual's age.
WORK	Individual's work status. 1: Full time worker, 2: Part-time worker, 3: Unemployed.
GRAD	Individual's educational attainment. 1: High school or less. 2: Undergraduate

	university studies. 3: Graduate or post-graduate degree.
RELIG	Individual’s religion. 1: No religion, 2: Christian, 3: Muslim, 4: Jewish.
STUDENT	Individual’s current student status. 1: Not a student, 2: University student, 3: Graduate school student.
MARRIED	Marital status. 1: Single, 2: Married, 3: Divorced.
PURCH	Have you ever purchased groceries online? 1: Yes, 2: No.
FACEBOOK	Are you a regular Facebook user? 1: Yes, 2: No.
TWITTER	Are you a regular Twitter user? 1: Yes, 2: No.
INSTA	Are you a regular Instagram user? 1: Yes, 2: No.
INCOME	Individual’s income in euros per year. 1: Less than 15,000, 2: Between 15,000 and 30,000, 3: More than 30,000
KIDS	Do you have kids? 1: Yes, 2: No.
ALONE	Do you live alone? 1: Yes, 2: No.
REGION	Individual’s residence region. 1: Grand Est, 2: Nouvelle-Aquitaine, 3: Auvergne Rhône-Alpes, 4: Bretagne, 5: Bourgogne France-Comté, 6: Centre Val de Loire, 7: Corse, 8: Languedoc-Roussillon Midi-Pyrénées, 9: Normandie, 10: Nord-Pas-de-Calais Picardie, 11:Île-de-France, 12: Pays de la Loire, and 13: Provence-Alpes-Côte d’Azur.
DENSIT	Residential density of individual’s neighborhood. 1: Urban, 2: Suburban, 3: Rural.
FREQ	How often do you drink champagne? 1: Occasional drinker, 2: Regular drinker.
DRINKER	How are you as a wine drinker? 1: Absolute beginner, 2: Fairly ignorant wine-lover, 3: fairly knowledgeable wine-lover, 4: expert.
PRICE_X	Price in euros of alternative X (X can be DEL, GOS, VCP, or HM).
KNOWN_X	Is brand X well known? 1: Yes, 2: No.
BOTTLE_X	Rate the style/design of brand X’s bottle.
INFLAB_X	Rate the informative label of brand X’s bottle.
LAB_X	Rate the style/design of brand X’s labels.
SMELL_X	Rate the smell of brand X’s champagne.
COLOR_X	Rate the color of brand X’s champagne.
ADV_X	Rate the advertisement (you have seen previously to this experiment) of brand X
UNO	Column full of 1’s